LOCALIZATIONS, ETALE MAPS AND FILTERED COLIMITS OF WEAKLY ETALE ALGEBRAS ARE WEAKLY ETALE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Weak etaleness of a ring map as flatness together with flatness of the multiplication B tensor_A B -> B. used at proof lines 30455-30456 and 30465-30468. Is the property being verified in each of the three cases.
- Permanence of weak etaleness and of flatness under composition and filtered colimits. used at proof lines 30474-30480. Transports flatness of the multiplication maps along the directed system.
- Etale maps between etale A-algebras are etale. invoked at proof lines 30466-30467. Gives flatness of the multiplication map in the etale case.

Source reference. Original lemma 05UT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7527–7527. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05UU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7528–7528. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00HC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 638–638. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00U7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1101–1101. Cross-chapter navigation pointer into

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Source reference. Original lemma 30405: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30405–30405. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $A \to B$ be a ring map. Then $A \to B$ is weakly étale in each of the following cases*

(1) *$B = S^{-1}A$ is a localization of A,*
(2) *$A \to B$ is étale,*
(3) *B is a filtered colimit of weakly étale A-algebras.*

Proof. An étale ring map is flat and the map $B \otimes_A B \to B$ is also étale as a map between étale A-algebras (Algebra, Lemma 00U7). This proves (2).

Let B_i be a directed system of weakly étale A-algebras. Then $B = \operatorname{colim} B_i$ is flat over A by Algebra, Lemma 05UT. Note that the transition maps $B_i \to B_{i'}$ are flat by Lemma 30405. Hence B is flat over B_i for each i, and we see that B is flat over $B_i \otimes_A B_i$ by Algebra, Lemma 00HC. Thus B is flat over $B \otimes_A B = \operatorname{colim} B_i \otimes_A B_i$ by Algebra, Lemma 05UU.

Part (1) can be proved directly, but also follows by combining (2) and (3). □